In-process QC Procedure

For

Onboard Kavach Installation In HHP Loco

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Abstract
This document describes the standard procedure of In-process QC Procedure for Onboard Kavach Installation

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1. Purpose

The purpose of this manual is to specify Onboard KAVACH equipment Quality Inspection procedure to inspect quality issues in Locomotives.

2. Scope

Scope of this document is to lay down the quality inspection procedure of onboard KAVACH and its sub-component units during installation and commissioning. Quality incharge shall be responsible for maintaining this Procedure.

3. Quality Check Points:

3.1 Onboard KAVACH:

3.1.1 Confirm that the location of the Loco Kavach unit in the loco allows access for easy replacement of PCBs.

3.1.2 Check the Loco Kavach unit stand welding without any gaps with respect to the loco surface.

3.1.3 Ensure that there is no loose joint between stand to surface during welding.

3.1.4 Ensure that finishing of the welding to be smooth and there is no sharp edges.



Figure 1: Loco KAVACH Stand Mounting

3.1.5 The welded loco stand shall be made of 5mm thick steel, to withstand Loco vibrations. The structure shall be stable and standing on four legs. Cantilever structures with two legs shall not be used.

3.1.6 The stand shall be treated with red oxide coating, before being painted with RAL7032 (Smoke Grey/Pebble Grey color) paint, to avoid corrosion.

3.1.7 Ensure that Loco stand fixing holes(M8) are exactly matching with Loco channel fixing holes(M8).

3.1.8 The M8 bolts shall be tightened with 25N-m torque, using a torque spanner. The bolt heads shall be marked with green paint, after the QC inspector checks for proper torque application.

3.1.9 Check the cable connections are made as per the labeling without overlaps and wrong connections.

3.1.10 Ensure that all the external cable circular connectors are fully locked properly with Loco Kavach receptacles.

Improper Cable Routing **Proper Cable Routing**



Figure 2: Loco Kavach unit cables routing

3.2 EMI Filter Box:

3.2.1 Ensure that fixing of EMI Filter Box to loco stand with (M5x16mm) screws and nuts with torque wrench and maintained torque 6N-m for proper tightness of fixing.

3.2.2 Check the cable routing and ensure that there are no sharp bends and stress on cables.

3.2.3 Ensure all the cables routed through PG glands and also check all the cables are tied with cable ties.

Proper Cable Routing

Improper routing of EMI filter cable

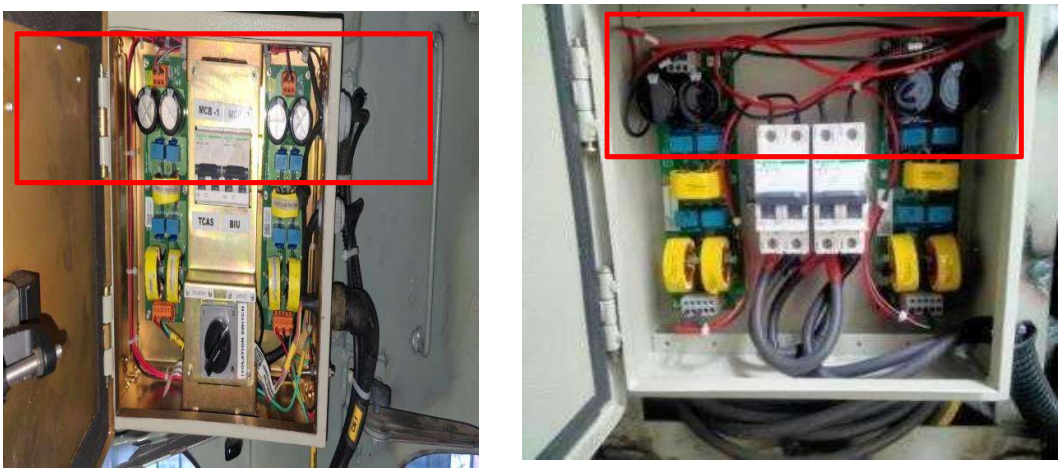


Figure 3: EMI Filter Unit Mounting

3.3 RIB and Cab Input Box:

3.3.1 Ensure that enough space available in-between cab input box and RIB unit for easy access of cables.

3.3.2 Check the RIB and CAB Input box stand welding should be proper without any gap on welding.

3.3.3 The welded loco stand shall be made of 5mm thick steel, to withstand Loco vibrations.

3.3.4 Welding portion to be clean and smooth without any cracks and joint breaks.

3.3.5 The stand shall be treated with red oxide coating, before being painted with RAL7032 (Smoke Grey/Pebble Grey color) paint, to avoid corrosion.

3.3.6 The M5X16mm bolts shall be tightened with 6N-m torque, using a torque spanner. The bolt heads shall be marked with green paint.



Figure 4: RIB & CAB I/P BOX Mounting

3.3.7 Cables to be routed without any sharp bends, overlaps and tied properly with cable ties.

3.3.8 Ensure that cables are connected with respect to labels and ensure circular connectors are fully locked with enclosure receptacles.



Figure 5: RIB & CAB I/P Cable Routing

3.4 DMI/LP-OCIP Mounting:

3.4.1 Check the DMI mounted place is good enough at driver desk, which can be operated easily by Loco pilot.

3.4.2 Ensure that DMI mounting frame is properly Ensure that mounting The M5X16mm screws shall be tightened with 6N-m torque, using a torque spanner. The screws heads shall be marked with green paint.



Figure 6: DMI/LP-OCIP Mounting stand

3.4.3 Ensure that cable routing done properly without over stress and sharp bends.

3.4.4 Ensure that the cable booting is not damaged/peeled out while or after routing the cable.

3.4.5 DMI-1 cable shall be connected to MC1 and DMI-2 cable shall be connected to MC3 at Loco Kavach unit.

3.4.7 Ensure that the circular connectors are fully locked with DMI unit enclosure receptacles.



Figure 7: DMI/LP-OCIP Cable Routing

3.5 RFID PS Box:

3.5.1 Ensure that cable connections are given at the corresponding location with respect to labels provided at Loco Kavach unit.

3.5.2 Ensure that cables are routed properly without hanging and tied properly with cable ties.

3.5.3 Ensure the circular connectors are properly locked with RFID box unit receptacles.



Figure 8: RFID PS Box Fixing and Cable Routing

3.6 Radio & GPS-GSM Antenna's:

3.6.1 Check the Radio antenna's base plate welding done properly without any joint gaps and cracks.

3.6.2 Before welding the RF and GPS-GSM antenna, height should be followed as mention in below table.

#	Item Description	Required Height. Max in (mm)
1	RF Antenna – TX1 from Base (From long-leg of antenna)	≤ 310
2	RF Antenna – TX2 from Base (From long-leg of antenna)	≤ 310
3	GPS-GSM Antenna1 from Base (From long-leg of antenna)	≤ 310
4	GPS-GSM Antenna2 from Base (From Long-leg of antenna)	≤ 310
5	RF Antenna height – from Rail Level	≤ 3960
6	GPS-GSM Antenna height – from Rail Level	≤ 3891



Figure 9: Radio(RF) Antenna Fixing

3.6.3 Ensure antenna's shall be fixed as per RDSO diagram and welding shall be done to withstand loco vibrations, while running with higher speeds.



Figure 10: GPS&GSM Antenna Fixing

3.6.4 Verify antenna's height as per diagram, ref:OHE line contact **RDSO Lr.No.EL/0.1.3/3**





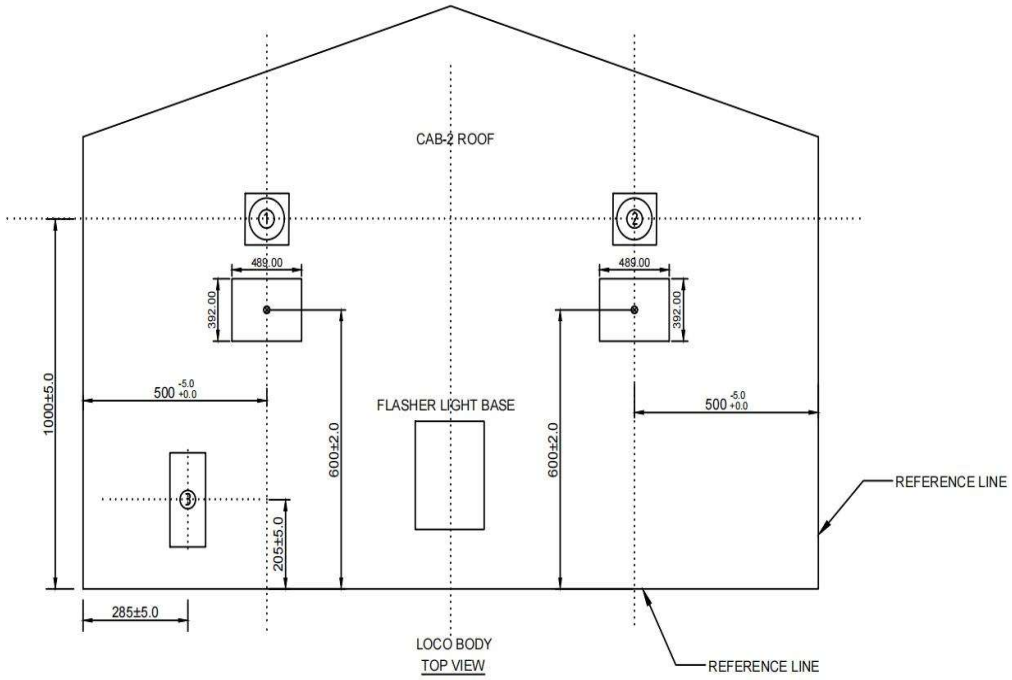


Figure 11: GPS&GSM,TX&RX Antenna Installation and Height Measurement

3.6.5 Ensure that the LMR400(TX-1,TX-2,RX-1&RX-2)&LMR200(GSM1, GPS1, GSM2& GPS2)cables are connected to their respective locations.

3.6.6 Ensure that the cables are routed properly through the 2" steel reinforced pipe/hose to avoid environmental damages and pipe/hose tied to supporting welded rods using cable ties.

3.6.7Ensure that cables are not exposed outside after routing through reinforced pipe/hose. Any locally procured conduit shall be of

KONTACT(or Equivalent) make and steel wire reinforced pipe model.

3.6.8 Ensure that the all welded portions shall be treated with red oxide coating, before being painted with RAL7032 (Smoke Grey/Pebble Grey color) paint, to avoid corrosion.

3.7 Pneumatic fittings and EPValve,Cocks fixing :

3.7.1 Confirm that all pipes and fittings used in the assembly are from the approved BOM, and taken from the I&C kit supplied from the factory.

3.7.2 Confirm that copper pipes are bent using appropriate bending tool, and that there no kinks or sharp bends in the pipe.

3.7.3 Ensure that copper tube length is measured with respect the connectivity from loco pneumatics to EP Valve, BP cock, Horn cock and valve arrangements.

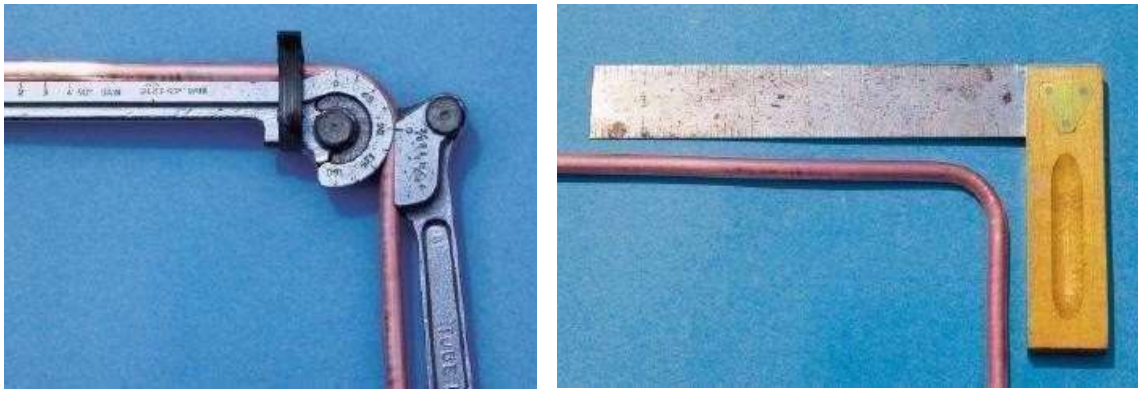


Figure 12: Copper pipe bending process with bend tool

3.7.4Ensure that copper pipe connections made properly with approved make(Ex.Fluid Control)ferrules and TEE-joints used.

3.7.5 All threaded connections must be sealed with Teflon paste loctite 567 (NOT A TEFLON TAPE), which is supplied in the I&C kit

3.8 Installation of Speedometer and Speedometer Sensor :



Figure 13: Speedometer Assembly

3.8.1 Ensure that speedometer fixing holes are matched with fixing clamps holes, uneven fixing may cause loose connection of speedo meter will leads to peel out from locomotive.

3.8.2 Ensure that cables from PG to Speedometers and from speedometer units to Loco Kavach unit are proper.

3.8.3 Identify the location of the axle where Speed Sensor to be installed.

3.8.4 Remove damper & axle box cover.

3.8.5 Ensure bearing end cap on the axle is removed and then the speed sensor end cap is mounted in its place as shown in picture.

3.8.6 Ensure speed sensor cover (dust cover) is placed over the speed sensor end cap and to ensure concentric of the end cover w.r.t speed sensor cap 3 radius gauges are placed between the speed sensor end cap and sensor cover as shown in below picture



Figure 14: PG's & Speedometer Sensor Assembly

3.9 RFID Reader Assembly:

3.9.1 Check and Ensure that RFID reader cables are routed properly through the loco trench without any overlaps and over stress.

3.9.2 Is mud guard (from the Loco Kavach Installation Kit) been fixed to RFID reader mounting bracket, installed in front of each RFID reader on the cattle guard side, using the supplied bolts and washers.

3.9.3 Ensure that cable shall not have any sharp bends in routing and also ensure that cables shall not have any damages while routing.

3.9.4 Ensure that mil connector booting was not damage in routin

3.9.5 Ensure that mill connectors connectivity is as per the wiring at loco side and RFID Reader side.

3.9.6 Ensure the cattle guard cutting of supporting brackets has been followed as per **File No RDSO-MP0LKO(EC)10/2020 .**

3.9.7 Ensure the cattle guard plate should be cut using cold cutting process.

3.9.8 Ensure for strengthen the cut plate, weld an additional 10 mm plate as shown in the sketch in **Annexure-1 .**

3.9.9 During installation, the bottom surface of the RFID reader shall be adjusted to be at a height of 400 ± 50mm from the rail head.

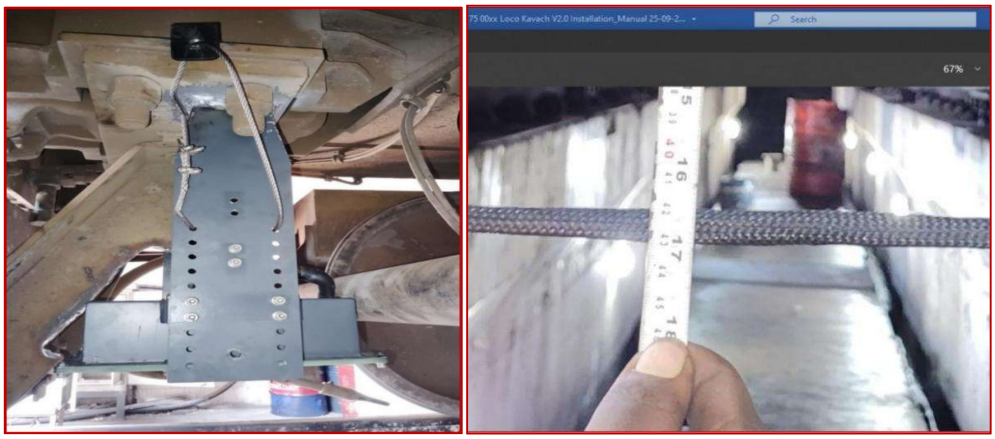


Figure 15: RFID Reader Installation& Height Measurement

3.9.10 Ensure that RFID Reader-1 is connected to MC6 and RFID Reader-2 connected to MC7 at Loco Kavach.

3.9.11 Ensure that the circular connectors are properly locked with RFID box unit receptacles.

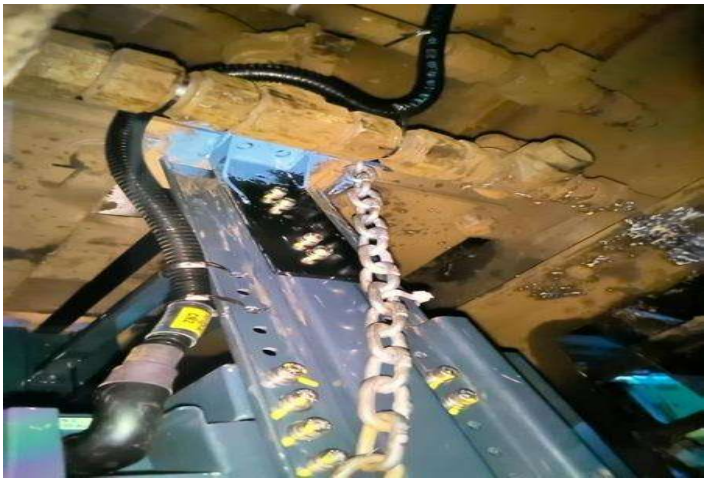


Figure 16: RFID Reader cable assembly and routing

3.10 Earthing:

3.10.1 Ensure that earthing done with 10Sq.mm Yellow/Green cable for Loco Kavach main unit, RIB unit, CAB Input Box, LP-OCIP units on both the cabs.

3.10.2 Ensure that cable continuity, lugs crimping and tightness of earth cable.

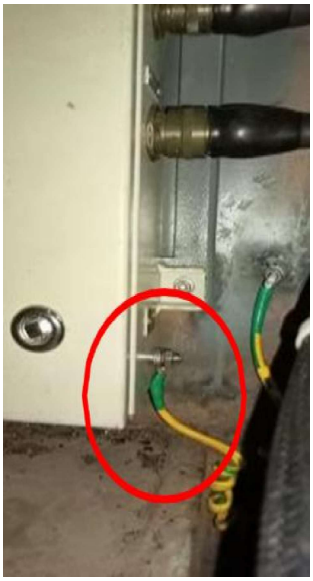
3.10.3 Ensure that the earth cables are routed through the conduit and routed properly and tied with cable ties.



Kavach Unit Earthing



RIB Earthing



CAB I/P Earthing



Isolation Box Earthing **Radio Unit Earthing** **DMI Earthing**

Figure 17: Earth cable connectivity for all units

3.11 Installation of CCB modules:

3.11.1 PSJB & TPM modules Installation:

3.11.2 The CCB Brake panel, already installed in the locomotive, has three sub-assemblies called Junction Box, PSJB Box and MPIO module, as shown in the following picture.



Figure 18: PSJB module Installation

3.11.3 The PSJB Box shall be replaced with a new one, if the already installed PSJB has a date code of 2020-03 or earlier. The date code is on the name plate fixed to the front of the PSJB box. The first four digits engraved in the last row, represent the year and month of manufacture. In the picture shown below, the PSJB was manufactured in Jan 2020, and so, shall have to be replaced with a new one.

3.11.4 If the PSJB Box has to be replaced, the original PSJB Box shall be removed, and the new PSJB Box (supplied in the Installation Kit) shall be

installed in its place.

3.11.5 TPM module, supplied in the Installation Kit shall be installed on the right side of the existing MPIO module as shown in the following figure.



Figure 19: PSJB & TPM module Installation

3.11.6 The removed modules shall be handed over to the Loco Shed officials.

3.11.7 Ensure the Isolation cock shall be fixed to SIFA valve for easy operation by loco pilots.

3.11.8 Ensure the copper pipes bending done with proper tools to avoid wrong bending or damages.

3.11.9 Ensure that SIFA VALVE SHOULD BE CLOSED CONDITION WHILE INSTALLATION.

3.11.10 Ensure the fixing of copper pipe using T-Joint(1/2”) at BP pipe line and Elbow (1/2”) at SIFA valve in port as shown in the below figure.

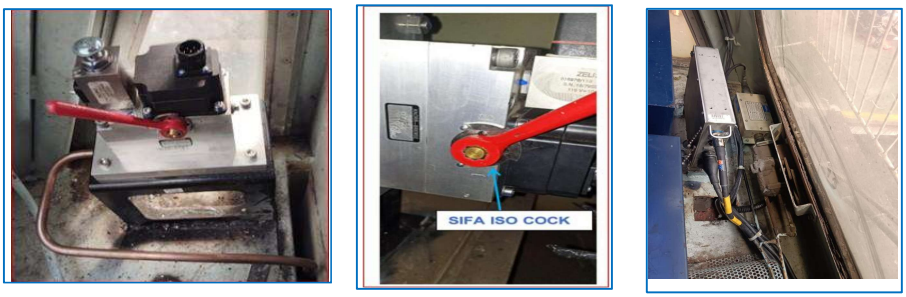


Figure 20: SIFA& TPM module Installation

3.12 Installation of Pressure Sensors (MR, BP, BC1) at CAB - I & BC 2 Pressure Sensor at CAB – 2

3.12.1 All these pressure sensors shall be installed under CAB-I driver desk except BC2 sensor. BC2pressure sensors shall be installed under CAB-2 driver desk.

3.12.2 T-Joint, supplied from factory, shall be installed on MR gauge pipe under the driver desk. 16-bar pressure sensor shall be installed on this T-joint.

3.12.3 T-Joint, supplied from factory, shall be installed on BP gauge pipe. 7-bar pressure sensor shall be installed on this T-joint.

3.12.4 T-Joints, supplied from factory, shall be installed on BC1 & BC2 gauge pipes. Two 7-bar pressure sensors shall be installed on these T-joints.

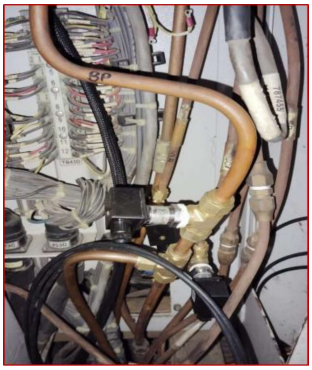
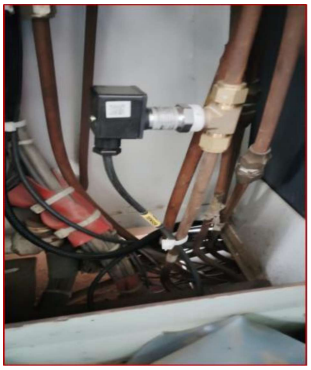



Figure 21: Pressure Sensors MR,BP,BC1&BC2

3.13 Instructions for fastening with bolts and nuts:

3.13.1 All bolts and nuts shall be secured with suitable spring washers and plain washers. These washers are already supplied in Loco Kavach Installation kit, separately packed for each sub-system to be installed.

3.13.2 All threaded connections (bolts, nuts, screws etc.,) shall be secured with thread-locker, supplied in the Loco Kavach Installation kit.

3.13.3 Suitable torque spanners shall be used to apply torque appropriate to each screw, bolt or nut, per the table below and mark with (Yellow/Green) Paint.

Fastener size	Torque applied, N-m
M5	5
M6	10
M8	25
M10	40
M12	76

4.0 QC Report:

4.1.1 During the installation of Loco Kavach, the QC Engineer shall ensure the quality of installation and update the results in the QC in-process checklist (**EG-IC-FT-34**).

4.1.2 If any observations are found, they shall be recorded in the report and escalated to the Shed In-charge for necessary action. After completion of the rework, re-verification shall be carried out, and the status of rework along with the date shall be updated in the audit report.